Day 2 Evening — Web Technologies

CSS selectors, CSS types, typography, div, box model, spacing and colors | Based on Abhishek's Oct 22 lecture transcript + your reference code

How to use these notes: Read in the lecture sequence, run each code example in VS Code, then use the practice tasks and quick revision section. Examples are cleaned up for readability.

1. CSS basics and syntax

CSS (Cascading Style Sheets) controls the presentation of HTML elements: text color, font, dimensions, borders, spacing and more. In internal CSS, write CSS rules inside a **<style>** element, normally inside the HTML **<head>**.

```
<style>
  selector {
    property: value;
  }
</style>
```

Selector = which HTML element(s) to target. **Property** = what visual feature to change. **Value** = how it should look.

2. Types of CSS selectors

A. Tag selector

Write the HTML tag name directly. It selects all matching elements, so one rule styles every matching tag.

```
<style>
  h1 { color: red; }
  p  { font-size: 20px; }
</style>
```

If a page has three h1 elements, all three become red. Use this when all elements of that tag should share a style.

B. ID selector

An **id** is intended to identify one element uniquely within a page. In CSS, prefix the id with **#**. The # symbol is part of the CSS selector, not part of the id attribute value.

```
<style>
  #head-1 { color: red; }
  #head-2 { color: green; }
  #head-3 { color: blue; }
</style>

<h1 id="head-1">Hello</h1>
<h1 id="head-2">Heading 2</h1>
<h1 id="head-3">Heading 3</h1>
```

Good naming: head-1, main-title, login-form. Avoid reusing the same id on multiple elements. Browsers may still render a page with duplicate IDs, but it is invalid HTML practice and can cause bugs.

C. Class selector

A **class** lets you group multiple elements that should receive common styling. Reusing a class is normal. Prefix the class name with a dot (**.**) in CSS; do not include the dot in the HTML class value.

```
<style>
  .blue-font { color: blue; }
</style>

<h1>Hello</h1>
<h1 class="blue-font">Heading 2</h1>
<h1>Heading 3</h1>
<p class="blue-font">para 1</p>
<p>para 2</p>
<span>span 1</span>
<span>span 2</span>
<span class="blue-font">span 3</span>
```

The second h1, first p and third span become blue. One class rule styles all three, even though they are different tag types.

D. Universal selector

The asterisk (*****) selects all elements. Use it carefully because it can affect every element.

```
* {
  box-sizing: border-box;
}
```

A common practical use is setting box-sizing consistently. For changing a specific property, a tag, class or id selector may be more suitable.

3. Selector priority and conflicting styles

The lecture introduces this simplified selector-priority order when the same property conflicts: **ID > class > tag > universal**.

```
*        { color: blue; }
h1       { color: green; }
.red-font { color: red; }
#head-1  { color: yellow; }

<h1 id="head-1" class="red-font">Heading</h1>
```

For the color property in this example, yellow wins because the ID selector is more specific. Other non-conflicting properties from the other rules can still apply: CSS does not discard the entire lower-priority rule.

Important accuracy note: The lecture's order is a useful beginner shortcut for these selectors, but the full CSS cascade also considers origin/importance, layers, specificity, and source order. Inline styles are not simply the next specificity level in every possible case, and !important rules can change the result. For now, first compare whether the rules set the *same property*, then compare specificity.

4. Three ways to write CSS

A. Internal CSS

CSS is written in a <style> element inside the same HTML file, usually in <head>. Useful for a single page or small demos.

```
<head>
  <style>
    p { color: purple; }
  </style>
</head>
```

B. External CSS

CSS is placed in a separate .css file. Link it to the HTML document from the head. This improves modularity and reusability: one stylesheet can be linked to many HTML pages.

```
<!-- index.html -->
<head>
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <h1>Heading</h1>
</body>

/* style.css */
h1 {
  color: green;
}
```

Attributes: rel="stylesheet" identifies the relationship; href="style.css" gives the file path. The lecture also demonstrates type="text/css"; modern HTML normally does not require this type attribute for CSS links.

If the CSS file is in a folder called css, a typical path is href="css/style.css". The path must match your project structure.

C. Inline CSS

CSS is written directly on an HTML element using the style attribute.

```
<p style="color: red;">This is para 1</p>
```

The lecture's simplified order for otherwise comparable conflicting declarations is: **inline > internal > external**. More precisely, CSS cascade rules decide the winner; stylesheet location alone does not universally determine priority. Inline styles commonly beat ordinary author stylesheet declarations, but !important and other cascade rules matter.

5. Font and text properties

```
p {
  font-size: 24px;
  font-family: "Segoe UI", Arial, sans-serif;
  font-weight: 900;
  font-style: italic;
  font-variant: small-caps;
  text-transform: capitalize;
  text-decoration: line-through;
}
```

font-size: size of text, e.g. 24px. **font-family:** typeface; fallback fonts can be listed after commas. **font-weight:** thickness/boldness (common values include normal, bold, or numbers such as 100–900 when supported). **font-style:** normal, italic or oblique. **font-variant: small-caps:** renders lowercase letters as smaller capital-style letters. **text-transform:** uppercase, lowercase or capitalize (capitalizes the first letter of each word). **text-decoration:** underline, overline or line-through.

These properties change how text is displayed; text-transform does not rewrite the original text stored in the HTML.

6. The div element

 is a generic block-level container used to group elements into sections or layout regions, such as a navigation area, content panel or footer. By default, a div has no special background, border or fixed dimensions. Its size depends on its content and normal layout rules; as a block element it typically takes the available width.

```
<div class="card">
  <h3>This is heading</h3>
  <p>This is dummy para.</p>
</div>

<style>
  .card {
    width: 300px;
    height: 300px;
    background-color: green;
    color: white;
    border: 10px solid black;
    border-radius: 20px;
  }
</style>
```

Use an id or class to style one div or a group of divs. For modern HTML, semantic elements such as <header>, <nav>, <main>, <section> and <footer> can be clearer when their meaning fits.

7. Width, height, background, border

width and **height** set the dimensions. **background-color** sets the background; **color** sets text color and is inherited by many child elements.

```
div {
  width: 300px;
  height: 300px;
  background-color: green;
  color: white;
  border-width: 10px;
  border-color: black;
  border-style: solid;
  border-radius: 20px;
}
```

The border can also be written as shorthand: border: 10px solid black;. Border styles include solid, dashed, dotted and double. Border-radius rounds corners.

8. Padding vs margin

Padding = internal space between the content and the element's border. **Margin = external space** outside the border, separating the element from neighboring elements.

```
.box {
  padding-left: 20px;
  padding-top: 10px;
  padding-right: 20px;
  padding-bottom: 10px;

  margin-left: 50px;
  margin-top: 10px;
}
```

Side-specific properties: padding-left, padding-top, padding-right, padding-bottom; similarly margin-left, margin-top, margin-right, margin-bottom.

Shorthand padding and margin

Four values are read clockwise: **top, right, bottom, left**. Two values mean **top/bottom, left/right**. One value applies to all four sides.

```
padding: 10px;             /* all four sides */
padding: 10px 20px;        /* top/bottom 10; left/right 20 */
padding: 10px 20px 15px 40px; /* top right bottom left */
```

```
margin: 10px 20px;         /* top/bottom 10; left/right 20 */
```

With three shorthand values, the order is top, left/right, bottom. Margin and padding are not text alignment: text-align: justify aligns text edges; it does not create padding.

9. Box model and box-sizing

The browser's box model includes content, padding, border and margin. By default, when you set a width/height, padding and border are added outside that content size, so the rendered outer size can grow.

```
.box {
  width: 300px;
  height: 300px;
  padding: 20px;
  border: 5px solid black;
  box-sizing: border-box;
}
```

With box-sizing: border-box, the declared width and height include content + padding + border, so adding padding/border does not increase the declared outer dimensions (assuming ordinary sizing conditions). Margin remains outside.

In browser Developer Tools, inspect an element and find the **box model** diagram to see content, padding, border and margin values.

10. Color values in CSS

The lecture covers three common ways: color names, hexadecimal values, and RGB. Color names are simple but offer a limited named set; hex and RGB let you choose more precise shades.

```
h1 { color: red; }                /* named color */
p  { color: #3b5998; }            /* hexadecimal */
div { background-color: rgb(0, 128, 255); }
```

Hex: typically #RRGGBB, with two hexadecimal digits each for red, green and blue. Valid hex digits are 0–9 and A–F; for example, #671t43 is invalid because t is not a hexadecimal digit.

RGB: rgb(red, green, blue); in the common integer form each channel ranges from 0 to 255. rgb(0, 0, 0) is black, rgb(255, 255, 255) is white, and mixing channel values creates other colors.

RGBA: the fourth value controls alpha/transparency, usually from 0 (fully transparent) to 1 (fully opaque), e.g. rgba(255, 0, 0, 0.5). This was mentioned at the end of the transcript.

11. Complete practice page

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>CSS Demo</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <h1 id="main-heading">CSS Practice</h1>
  <p class="blue-font">This paragraph uses a class.</p>
  <div class="card">
    <h3>A small card</h3>
    <p>Practice padding, margin and borders.</p>
  </div>
</body>
</html>

/* style.css */
* {
  box-sizing: border-box;
}
#main-heading {
  color: #3b5998;
}
.blue-font {
  color: blue;
  font-size: 20px;
  text-transform: capitalize;
}
.card {
  width: 300px;
  min-height: 180px;
  padding: 20px;
  margin: 20px 0;
  color: white;
```

```
    background-color: rgb(0, 128, 0);
    border: 3px solid black;
    border-radius: 20px;
}
```

Save the HTML and CSS files in the same folder as index.html and style.css. Open the HTML page in a browser. Change one value at a time and observe the result.

12. Common confusions to remember

• **HTML id vs CSS ID selector:** HTML uses id="main-heading"; CSS uses #main-heading.

• **HTML class vs CSS class selector:** HTML uses class="blue-font"; CSS uses .blue-font.

• **style tag vs style attribute:** <style> contains internal CSS rules; style="color:red" is inline CSS.

• **External CSS:** a separate file is not automatically applied; the HTML must link to it with the correct href path.

• **color vs background-color:** color changes text; background-color changes the element's background.

• **Padding vs margin:** padding is inside the border; margin is outside it.

• **Box sizing:** border-box includes padding and border in the declared width/height; margin stays outside.

• **Selector priority:** when two rules set different properties, both may apply. Priority matters when declarations conflict on the same property.

13. Practice exercises

Try without looking at the solutions:

■ Write CSS that makes every paragraph 18px.

■ Give only the h1 with id="welcome" a green color.

■ Make three unrelated elements blue using one shared class.

■ Use the universal selector to apply box-sizing: border-box.

■ Write an external CSS link for a file stored in a folder named css.

■ Make a paragraph uppercase and underline it.

■ Create a 250px × 150px div with a blue background, white text and rounded corners.

■ Explain the difference between padding: 10px 20px and margin: 10px 20px.

■ For padding: 1px 2px 3px 4px, identify each side's value.

■ Write an RGBA value for red with 50% opacity.

14. Quick revision checklist

■ Tag selector = tag name; ID selector = #id; class selector = .class; universal selector = *.

■ Use a unique id for one element; reuse classes for shared styling.

■ Internal = style tag in HTML; external = separate .css file linked with link; inline = style attribute.

■ External CSS enables modularity and reuse across pages.

■ Font properties: font-size, font-family, font-weight, font-style, font-variant.

■ Text properties: text-transform and text-decoration.

■ div groups elements into a block-level container.

■ Border shorthand: width, style, color; border-radius rounds corners.

■ Padding is internal; margin is external.

■ Four-value shorthand order is top, right, bottom, left.

■ Box model = content + padding + border + margin.

■ Colors: named, #RRGGBB, rgb(), rgba().

15. Self-test (answers at the end)

• Which symbol starts an ID selector?

• Which symbol starts a class selector?

• Which selector targets all elements?

• Which type of CSS is written in a separate .css file?

• Which HTML element links an external stylesheet?

• Which property changes text color?

• Which property creates internal space?

• In four-value padding shorthand, what does the second value mean?

• What does box-sizing: border-box change?

• What is the valid range of each RGB integer channel?

Answers: 1) # 2) . 3) * 4) External CSS 5) <link> 6) color 7) padding 8) right 9) Includes padding and border within the declared width/height 10) 0–255.